Curriculum Vitae/Resume

Seunghun Oh

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Education

University of California, Los Angeles (UCLA) *Starting Sep 2026*
*Incoming PhD Student in **Civil and Environmental Engineering***

Seoul National University (SNU) *Mar 2020 - Aug 2026*
*BS in **Mechanical Engineering**, Double Major in **Computer Science***

- **GPA: 3.94/4.0**, **Major GPA: 3.97/4.0** *5th of 92, Mechanical Engineering, SNU*
- 21 months of military service included

Gyeonggi Science High School for the Gifted (GSHS) *Mar 2017 - Feb 2020*
*Major in **Physics** and **Mathematics***

- 1 Year Early Entrance

Research Interests

- Trustworthy robot learning and mobile autonomy for navigation, autonomous driving, and human-centered robotics
- Robot perception and 3D spatial intelligence for SLAM, reconstruction, and multi-sensor fusion

Publications and Preprints

- **S. Oh**, C. Han and C. Tang*
  "CleanGuide: Clean-Action-Aligned Guidance for Deployment-Time Adaptation of Diffusion Policies"
  IEEE International Conference on Robotics and Automation (ICRA), under review, 2027.
- D. Zhu, **S. Oh**, R. Huang, Z. Huang, J. Ma, and C. Tang*
  "SCAPE: Scenario-Conditioned Simulation-Augmented Policy Evaluation"
  Conference on Robot Learning (CoRL), 2026. arXiv:2608.19425 [Project Page]
- **S. Oh**, J. Choe, D. Lee, D. Lee, S. Jeong, Y.-C. F. Wang, and J. Park*
  "SVRecon: Sparse Voxel Rasterization for Surface Reconstruction"
  arXiv preprint. arXiv:2511.17364, 2025. [Project Page]
- H. Song, D. Lee, **S. Oh**, M. Jung, and A. Kim*
  "The City That Never Settles: Simulation-based LiDAR Dataset for Long-term Place Recognition under Extreme Structural Changes"
  IEEE International Conference on Robotics and Automation (ICRA) Workshop, 2025.
  Best Research Award. arXiv:2505.05076
- **S. Oh**†, Y. Kim†, C. Song, and A. Kim*
  "LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns"
  Journal of the Korean Robotics Society (KROS), June 2025. KROS Journal Link
- S. Hahn†, **S. Oh**†, M. Jung, A. Kim, and S. Jung*
  "Quantitative 3D Map Accuracy Evaluation Hardware and Algorithm for LiDAR (-Inertial) SLAM"
  International Conference on Control, Automation and Systems (ICCAS), 2024. arXiv:2408.09727

† Co-first authors. * Corresponding author.

Patents

- D. Lee, J. Lee, W. Kim, **S. Oh**, and J. Park*
  "Method and Apparatus for 3D Reconstruction and Articulated Structure Estimation from Rest-State Multi-View Images" *Korean patent application in preparation.*

Research Experiences

Incoming Ph.D. Student, UCLA TAMS Lab (Advisor: Prof. Chen Tang) *Los Angeles, CA, USA*
University of California, Los Angeles *Feb 2026 – Present*

- **Researching inference-time safety guidance for diffusion policies**
  Developing residual-reward guidance methods that adapt diffusion-policy action chunks at inference time while preserving the original task behavior.
- **Developed SCAPE for scenario-conditioned policy evaluation**
  Built a simulation-augmented evaluation framework that combines limited real-world rollouts with large-scale simulation to predict scenario-specific robot policy performance.

Undergraduate Researcher, SNU-VGI Lab (Advisor: Prof. Jaesik Park) *Seoul, Korea*
Seoul National University *Sep 2025 – Feb 2026*

- **Developed a novel "Rest-State" framework for Articulated 3D Reconstruction**
  Contributed ideas to the core pipeline design and ran baseline experiments on large-scale scenes.
- **Advanced SDF-based Sparse Voxel Rasterization with 3D Foundation Models**
  Developed *SVRecon*, an SDF-based sparse voxel rasterization framework using Pi3-based initialization and hierarchical smoothness losses, achieving cleaner surfaces and faster convergence than SVRaster.

Undergraduate Researcher, RPM Robotics Lab (Advisor: Prof. Ayoung Kim) *Seoul, Korea*
Seoul National University *Dec 2023 – Aug 2025*

- **3D Reconstruction with SDF-based Sparse Voxel Rasterization**
  Developed an SDF-based voxel rasterization pipeline to enhance mesh accuracy, extending SVRaster's architecture. Focused on enforcing voxel-wise continuity using CUDA and PyTorch. → Open-source project
- **Long-term Place Recognition using Simulated LiDAR Datasets**
  Adapted and implemented long-term loop closure algorithms (Scan Context, BTC, Solid) in simulated urban environments. Evaluated PR performance under structural changes with customized pipelines.
- **LiDAR-Camera Calibration via Circular Marker Detection**
  Developed a robust LiDAR-based 6-DoF pose estimator using circular patterns, later integrated with *DiscoCal (CVPR)* into a full LiDAR-camera calibration system. → Open-source project
- **Quantitative Evaluation of 3D SLAM Map Accuracy**
  Designed experimental setups and metrics for evaluating LiDAR(-inertial) SLAM accuracy. Developed benchmarking tools for 3D map reconstruction quality comparison. → Open-source project

Research Student, Aerospace Engineering Club (Bulnabi) *Seoul, Korea*
Seoul National University *Dec 2023 – Apr 2024*

- Built foundational robotics environment using Ubuntu and ROS
- Simulated drone operations with Gazebo and QGroundControl

Research Student, Mechanical Engineering Club (ZERO) *Seoul, Korea*
Seoul National University *Mar 2020 – Aug 2020*

- Developed low-level C/C++ control modules in a Linux-based robotics environment.

Selected Awards and Honors

Scholarships / Fellowships

- **Kwanjeong Overseas Scholarship**, Kwanjeong Educational Foundation Fall 2026 – Spring 2030
  Annual scholarship of $25,000–$30,000 supporting Ph.D. research at UCLA.
- **The National Presidential Science Scholarship**, Korea Student Aid Foundation 2024 – 2025
  Full tuition plus approx. $3,700/year stipend for **four semesters**, awarded by the Korean government to top STEM undergraduates.
- **Work-Study Scholarship** (*Value Exploration and Practice*), Seoul National University Mar 2025 – Nov 2025
  Work-study research scholarship supporting an independent undergraduate project (approx. $1,500).
- **Work-Study Scholarship** (Type 1), Seoul National University Mar 2024 – Nov 2024
  Work-study research scholarship supporting an independent undergraduate project (approx. $2,200).
- **Merit-based Scholarship**, Dept. of Mechanical Engineering, Seoul National University Spring – Fall 2021

Awards and Honors

- 2026 – IPIU **Outstanding Paper Award**, Encouragement Prize, for "Sparse Voxel Rasterization for 3D Surface Reconstruction from Images" (selected for **oral presentation**)
- 2025 – Co-authored ICRA 2025 workshop paper [1]; awarded **Best Research Award** (1st Place), IEEE ICRA Workshop on Future of Construction
- 2025 – **Outstanding Undergraduate Research Opportunities Program Award**, Dept. of Mechanical Engineering, Seoul National University, selected **1 of 3 awardees** from a **57-student** cohort (top ∼5%)
- 2024 – **Grand Prize**, Mechatronics Competition
- 2022 – **Winner**, Startup Camp hosted by Hanbat University (During Military Service)
- 2021 – Academic Honor: Highest GPA in 2nd Semester (4.3/4.3)
- 2018 – **Training Lineup for IPhO (International Physics Olympiad)**
- 2018 – **Winner**, SNU Youth Engineering Frontier Camp

Presentations / Talks

- Poster Presentation, "LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns" Presented at KROS Spring Conference, Gangwon, Korea, 2025
- Seminar Talks, "Introduction to Linux","Drone software & simulation" "Introduction to ROS" link
  Robotics Club Seminar, Seoul National University, 2024

Leadership / Extracurricular Activities

Republic of Korea Air Force (ROKAF) Seoul, Korea
Sergeant, 15th Air Force Base, 256th Squadron Mar 2022 – Dec 2023

- Led a squad of 10 airmen, overseeing daily operations, safety, and training for unit readiness.
- Managed Confidential Level II documents and coordinated flight schedules with pilots.

Mentoring Volunteer Program Andong, Korea
Volunteer Instructor and Team Leader Jan 2025 – Feb 2025

- Served as volunteer instructor and mentor: taught physics classes, led Sports Day activities, and provided interview coaching and STEM mentoring for high school students in a one-week program.

Skills

C++, C, Java, Python, MATLAB, RISC-V Assembly, Bluespec, PyTorch, CUDA, ROS1/2, Arduino, Fusion 360, LaTeX